A6.1 Definition of the ordering model

The AXI4 protocol supports an ordering model based on the use of the AXI ID transaction identifier.

The principles are that for transactions with the same ID:

- transactions to any single peripheral device, must arrive at the peripheral in the order in which they are issued, regardless of the addresses of the transactions
- memory transactions that use the same, or overlapping, addresses must arrive at the memory in the order in which they are issued.

Note

In an AXI system with multiple masters, the AXI IDs used for the ordering model include the infrastructure IDs, that identify each master uniquely. This means the ordering model applies independently to each master in the system.

The AXI ordering model also requires that all transactions with the same ID in the same direction must provide their responses in the order in which they are issued.

Because the read and write address channels are independent, if an ordering relationship is required between two transactions with the same ID that are in different directions, then a master must wait to receive a response to the first transaction before issuing the second transaction.

If a master issues a transaction in one direction before it has received a response to an earlier transaction in the opposite direction then there are no ordering guarantees between the two transactions.

Note

Where guaranteed ordering requires a response to an earlier transaction, a master must ensure it has received a response from an appropriate point in the system. A response from an intermediate AXI component cannot guarantee ordering with respect to components that are downstream of the intermediate buffer. For more information see *Use of Device memory types* on page A4-73.

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